

Naming alcohols

To name an alcohol:

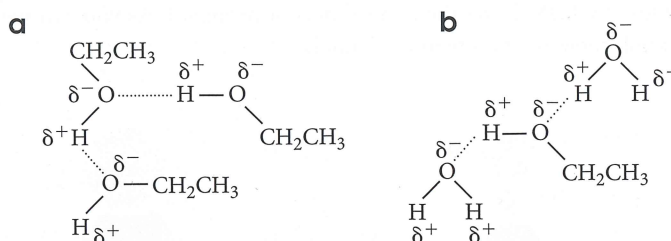
- 1 identify the longest carbon chain containing the —OH group
- 2 delete the 'e' of the parent alkane and add -ol; for example, ethane becomes ethanol
- 3 number the carbon atoms of the main chain from the end that produces the smallest number for the —OH group
- 4 add a number to the prefix to denote the position of the —OH group; for example, 2-butanol
- 5 name any alkyl groups as for branched alkanes.

Properties of alcohols

The presence of the highly electronegative oxygen atom means the C—O and O—H bonds are very polar. This means that hydrogen bonding can occur in alcohols. These strong intermolecular forces mean that alcohols have a higher melting and boiling point compared to their corresponding alkanes and alkenes.

The high polarity also means that the —OH group can form hydrogen bonds with water so the simpler alcohols are readily soluble in water, unlike alkanes and alkenes. However, as the carbon chain becomes longer, the molecule becomes more non-polar because the COH part becomes small compared to the longer alkyl chain, so solubility in water decreases with increasing chain length.

Figure C6.19 ►
Hydrogen bonding: a) in liquid ethanol, and b) between ethanol and water



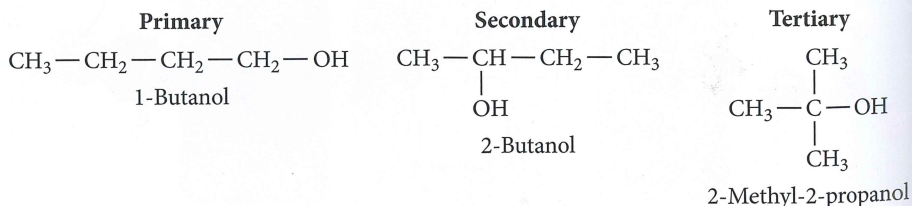
Types of alcohols

Alcohols can be classified according to the number of carbon atoms attached to the carbon bearing the —OH group.

- A **primary alcohol** is one in which one carbon atom is bonded to the carbon atoms attached to the —OH group.
- A **secondary alcohol** is one in which two carbon atoms are bonded to the carbon attached to the —OH group.
- A **tertiary alcohol** is one in which three carbon atoms are bonded to the carbon attached to the —OH group.

Figure C6.20 shows primary, secondary and tertiary alcohols with the molecular formula C_4H_9OH .

Figure C6.20 ►
Primary, secondary and tertiary alcohols



The three different groups of alcohols have different physical properties and chemical reactivity. This is because the strength of the hydrogen bonding in alcohols depends on the extent to which the —OH group is available for bonding. When the —OH group is on the end of the carbon chain, as in primary alcohols, it is more available for bonding than when it is surrounded by alkyl groups in tertiary alcohols. This means that the boiling points decrease from primary to secondary to tertiary alcohols, as can be seen in Table C6.6.

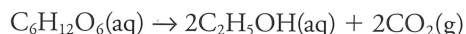
This also applies to their chemical reactivity. The most vigorous reaction with sodium occurs with a primary alcohol while the least vigorous is with the tertiary alcohol. Similarly, primary

alcohols are more readily oxidised by a strong oxidising agent such as potassium permanganate than secondary alcohols, whereas tertiary alcohols are not oxidised.

Alcohols contain the hydroxyl (—OH) functional group and can be classified as primary, secondary or tertiary. Their properties and reactivity are related to their classification.

Preparation of alcohols

Alcohols can be prepared in a number of ways. One of the most common ways of obtaining ethanol is through **fermentation**. In this process, yeast organisms consume sugars present in fruits such as grapes, sugar cane and grains, and produce ethanol. The equation for the fermentation of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is:



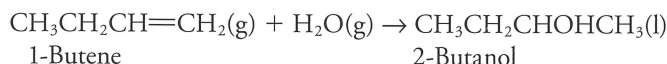
Methanol and blindness

Methanol is toxic to humans so high concentrations are lethal. Methanol is a by-product of fermentation with more produced in fruit fermentation than in grains due to the presence of pectins. Typically, there will be a small amount in wines. Higher methanol concentrations may be produced if fermenting occurs at high temperatures or a pectin enzyme is added. Professional winemakers monitor methanol concentrations to ensure they remain low. This is because the body's metabolic pathways that break down ethanol to make it harmless turn methanol into methanal, which is metabolised into methanoic acid. The cells of the optic nerve are extremely sensitive to methanoic acid, which is why blindness is closely associated with methanol poisoning.

Ethanol is not the only widely used alcohol, so a number of processes are used to synthesise alcohols.

The most common commercial method is the hydration of an alkene. This is an addition reaction. In the process, an alkene reacts with water in the presence of an acid such as sulfuric acid to produce an alcohol.

For example:

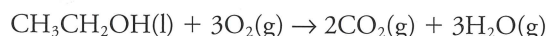


Reactions of alcohols

Combustion

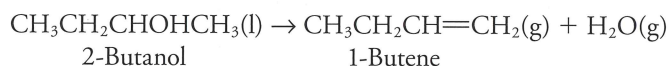
Alcohols undergo combustion. They burn readily in air to produce carbon dioxide and water.

For example:



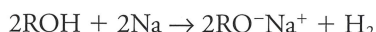
Dehydration

Alcohols also undergo dehydration by losing water to form alkenes when heated with concentrated sulfuric or phosphoric acids. (This is the reverse of alcohol preparation.) For example:

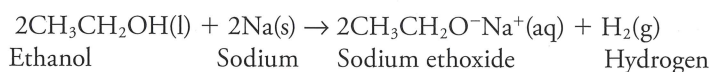


Reaction with sodium

Alcohols react with sodium to form an **alkoxide** anion (RO^-) and hydrogen gas. This is a similar reaction to the one between sodium and water but is less vigorous.



For example:



In this reaction, the alcohol acts as an acid. The alkoxide produced is a strong base and is the conjugate of the weak acid, which is the alcohol. The alkoxide readily reacts with water to form the original alcohol and produce OH^- ions.



To see more about fermentation in wine making, refer to Context 1, 'Wine', page 5.

When H—OH gets added across a carbon-carbon double bond, the H nearly always gets attached to the carbon with the most hydrogens already attached to it.

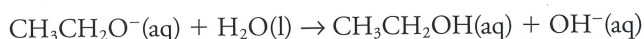
2-Butene is also formed in the dehydration reaction. Which isomer gets formed is a matter of chance.



MAIN REACTIONS OF ALCOHOLS

Visit this website to understand more about alcohols.

For example:



Oxidation

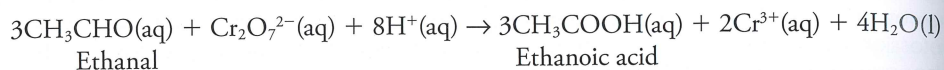
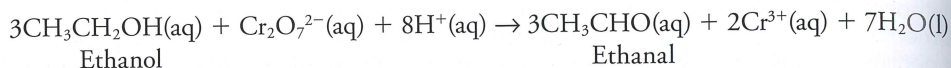
Alcohols readily undergo oxidation with strong oxidising agents such as acidified permanganate or dichromate solutions.

- a Primary alcohols are oxidised to form aldehydes, which are in turn oxidised to form carboxylic acids. Table C6.5 (page 297) shows the aldehyde and carboxylic acid functional groups.

These general reactions can be shown below. The [O] above the arrow in the equation indicates an oxidation process has occurred.



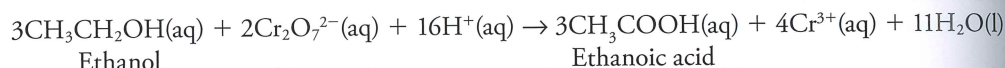
The following example shows the oxidation of ethanol to the aldehyde ethanal, then the further oxidation of ethanal to ethanoic acid using acidified potassium dichromate as the oxidising agent. (The oxidation and reduction half-reactions can be derived from the overall reactions given.)



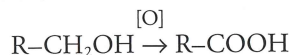
Aldehydes are easily oxidised to form carboxylic acids so the final product will not contain any aldehyde; it will contain only the carboxylic acid. A sample of an aldehyde can be obtained if an excess of the alcohol is used and the aldehyde is distilled off as soon as it forms.

The excess of the alcohol means that there is not enough oxidising agent present to carry out the second stage. Removing the aldehyde as soon as it is formed means that it does not get oxidised.

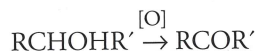
The overall equation for the oxidation of ethanol to ethanoic acid can be obtained by adding the two equations above together:



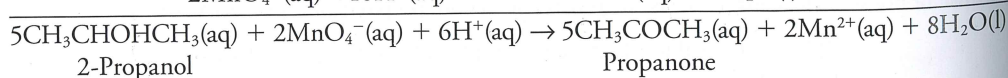
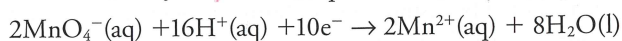
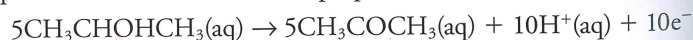
This can be represented generally as:



- b Secondary alcohols are oxidised to form ketones, which cannot be further oxidised. So the final product in the oxidation of a secondary alcohol is a ketone. (The ketone functional group is shown in Table C6.5.) Acidified permanganate and dichromate both oxidise secondary alcohols. The general equation for the oxidation of a secondary alcohol is shown below.



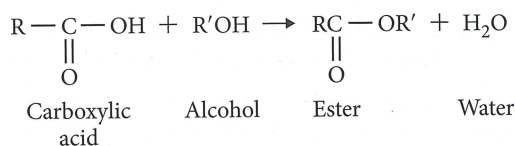
The half and overall equations for the oxidation of 2-propanol are:



- c Tertiary alcohols are not oxidised unless the conditions are drastic because this would involve a large amount of energy to break a C—C bond.

When dichromate is used as the oxidising agent, the orange $\text{Cr}_2\text{O}_7^{2-}$ ion is reduced to green Cr^{3+} . This test can be used to identify the presence of primary or secondary alcohols or aldehydes. With a tertiary alcohol or a ketone, there would be no change.

Alcohols react with carboxylic acids to form esters. The general equation for this reaction is shown below. These reactions will be discussed later in this chapter.



To see how oxidation of alcohol affects wine, refer to Context 1, 'Wine', page 21.



To learn about polymers based on alcohols, refer to Context 4, 'Polymer chemistry', page 92.



REVIEW REACTIONS

Visit this website to revise the oxidation of alcohols.

EXPERIMENT 6.2

DISTINGUISHING BETWEEN PRIMARY, SECONDARY AND TERTIARY ALCOHOLS (TEACHER DEMONSTRATION)

You will investigate the reactions of primary, secondary and tertiary alcohols with an oxidizing agent and with sodium to identify a way of distinguishing between the three groups.

Aim

To examine and compare the reactions of primary, secondary and tertiary alcohols

Materials

- 10 mL of ethanol, 1-butanol, 2-butanol, 2-methyl-2-propanol
- 5 mL of 0.1 mol L⁻¹ sodium dichromate (Na₂Cr₂O₇) solution
- 5 mL of 0.01 mol L⁻¹ potassium permanganate (KMnO₄) solution
- 5 mL of 6 mol L⁻¹ H₂SO₄
- 4 very small pieces of Na
- Hot plate or water bath
- Thermometer
- 250 mL beaker
- 5 large test tubes
- 10 mL measuring cylinder
- Metal tongs
- Dropper
- Tweezers

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Sodium dichromate is toxic.	Avoid contact with skin and inhalation of vapours. Wear gloves and a laboratory coat. Use in a fume hood. Wash your hands at the end of the experiment.
Sodium metal oxidises rapidly in air and is highly reactive with water. Contact will burn the skin.	Wear gloves and a laboratory coat. Do not allow contact with water to occur. Handle sodium only with tweezers and use only a very small amount.
Sulfuric acid is corrosive.	Wear gloves and personnel protective equipment. Avoid contact with the skin. Wash your hands at the end of the experiment.
Alcohols are highly flammable.	Ensure no flame sources are present in the location of these chemicals.

In your write-up, add any more risks you can think of, as well as ways to manage them.

Procedure

Part A Reactions with oxidising agents

Reaction with permanganate (MnO₄⁻)

- 1 Pour 2 mL of each of the alcohols into separate, labeled test tubes.
- 2 In a separate test tube mix 4 mL of potassium permanganate solution with 1 mL of sulfuric acid.
- 3 Using the dropper place 4–5 drops of the permanganate mixture into each of the test tubes containing the alcohols.
- 4 Record your observations, including smell.
- 5 Dispose of the solutions in a chemical waste jar as directed by your teacher. Do not pour them down the sink.

Reaction with dichromate (Cr₂O₇²⁻)

- 1 Prepare a water bath at a temperature of 80°C.
- 2 Pour approximately 2 mL of each of the alcohols into separate, labeled test tubes.
- 3 In a separate test tube mix 4 mL of sodium dichromate solution with 2 mL of sulfuric acid.
- 4 Place 1 mL of this solution into each of the test tubes containing the alcohols, shaking gently to mix.

- Place the test tubes in the water bath for 5 minutes. Shake gently occasionally.
- Record any changes including smell.
- Dispose of the solutions in a chemical waste jar as directed by your teacher. Do not pour them down the sink.

Part B Reaction with sodium metal

- Pour 2 mL of each of the alcohols into separate, dry, labelled test tubes.
- Obtain four very small pieces of sodium from your teacher. Place one piece in each test tube.
- Carefully observe the rates at which the reactions occur. Record your observations.
- Dispose of the solutions in a chemical waste jar as directed by your teacher. Do not pour them down the sink.

Results

Draw up a table to record your observations.

Analysis of results

- Rank the alcohols in order of reactivity.
- Generalise this to a statement about the relative reactivities of primary, secondary and tertiary alcohols.
- Write the names and formulas of the products formed where reactions took place.
- Write separate balanced equations for the oxidation of ethanol by MnO_4^- to ethanal and ethanoic acid.
- Write a general equation for the reaction of alcohols with sodium.

Conclusion

State in your own words the difference between the reactivities of the different classes of alcohols.

Extension

Use your understanding of the reactions of ethanol to use these tests to distinguish between alcoholic and non-alcoholic wine.

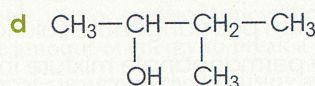
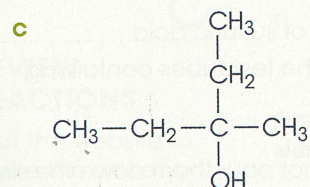
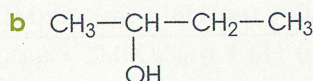
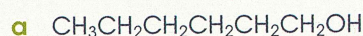
QUESTION SET 6.6

Remembering

- Name and give the structure of the alcohol functional group.
- Explain why:
 - alcohols are soluble in water.
 - solubility decreases as the alkane chain of the alcohol gets longer.
- Describe the three classes of alcohols and give an example of each class.

Understanding

- Explain how and why the properties of alcohols vary with the class of alcohol.
- Name the following compounds.



- Draw the structural formula of:
 - 2-propanol.
 - 3-methyl-1-butanol.
 - 2,3-dimethyl-3-pentanol.
 - 3-methyl-4-octanol.

Applying

- 7 Name and draw the structural formulas of two isomers of an alcohol with the molecular formula C_4H_9OH .
- 8 Write equations for the following reactions, naming the products.
 - a Dehydration of 3-pentanol
 - b Hydration of 1-propene
 - c Combustion of 2-butanol
 - d Reaction with sodium of 1-hexanol
- 9 Identify the final product (if any) from oxidising:
 - a 2-butanol.
 - b methanol.
 - c 2-methyl-2-hexanol.
- 10 Write balanced equations for the oxidation of:
 - a 1-propanol to propanal by acidified permanganate solution.
 - b 2-pentanol by acidified dichromate solution.

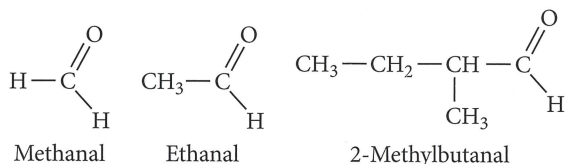
6.7 Aldehydes and ketones

Compounds containing a carbon atom double bonded to an oxygen atom are called **carbonyl compounds**. Both aldehydes and ketones are carbonyl compounds.

Aldehydes are carbon compounds that contain a carbon–oxygen double bond at the end of the carbon chain. The compound is represented by the general formula:

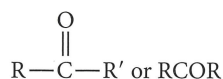


Examples of aldehydes are:

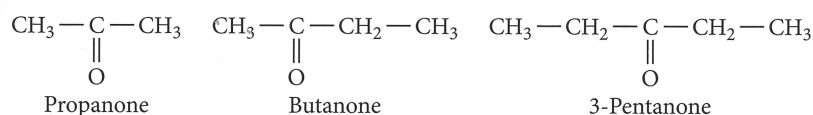


Methanal, HCOH , commonly known as formaldehyde, is the simplest aldehyde. Methanal is used in vast quantities in the manufacture of plastics. The first plastics, known as Bakelite, were made by using phenol and formaldehyde.

Ketones differ from aldehydes in that the $\text{C}=\text{O}$ can be located on any carbon except those at the ends of the hydrocarbon chain. The compounds have the general formula:



Examples of ketones are:



Propanone is widely used as a solvent and is commonly known as acetone.